

St. Francis Institute of Technology
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Department of Information Technology

A.Y. 2025-2026

Class: BE-IT A/B, Semester: VIII

Subject: Cloud Computing Lab

Student Name: Riya Sandip Indap

Student Roll No: 21

Experiment – 9: Implement Containerization

Aim: To study and Implement Containerization using Docker on AWS/Azure/Google cloud platform.(Free Tier Cloud Platforms).

Objective: After performing the experiment, the students will be able to learn –

- How to implement containerization using docker

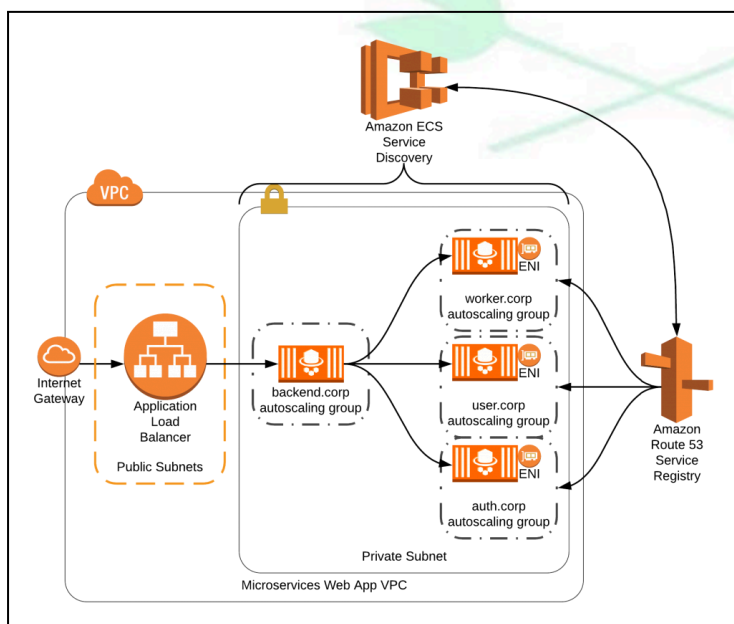
Lab objective mapped : ITDO8024.5 : To comprehend Docker and its role in the DevOps lifecycle.

Prerequisite: An AWS account

Requirements: Cloud Login, Desktop, Browser, Internet

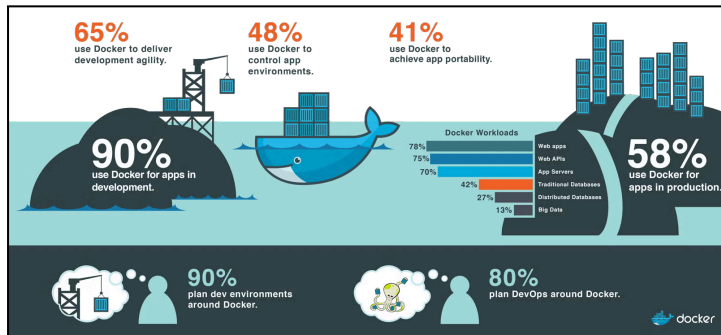
Pre-Experiment Theory: (describe following topics in soft copy form)

1. Amazon Elastic Container Service (Amazon ECS)



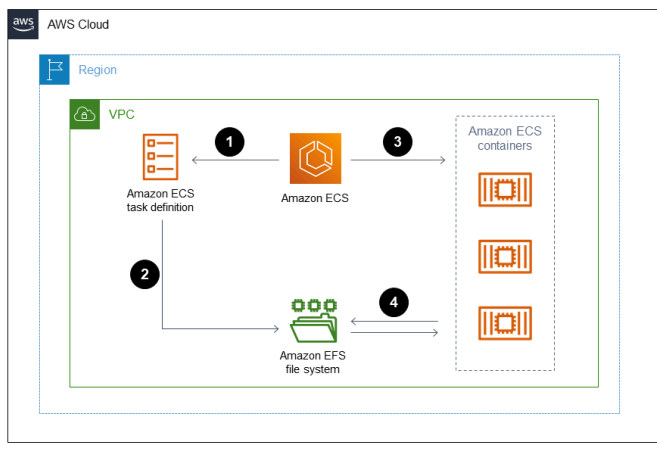
Amazon Elastic Container Service (Amazon ECS) is a fully managed container orchestration service provided by Amazon Web Services that allows users to run, manage, and scale containerized applications. It eliminates the need to install and operate container orchestration software by handling tasks such as scheduling containers, managing clusters, and ensuring application availability. ECS integrates with other AWS services like load balancers, security systems, and monitoring tools, making it easier to deploy reliable and scalable applications in the cloud.

2. Docker applications



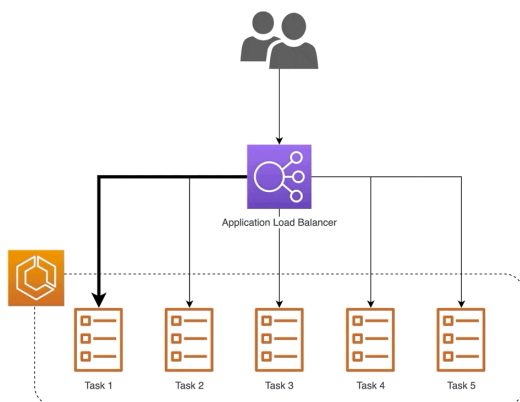
Docker applications are software applications that are packaged inside containers using Docker. These containers include everything needed to run the application, such as code, libraries, and dependencies, ensuring consistency across different environments (development, testing, and production). Docker applications are lightweight, portable, and can run on any system that supports Docker, which makes deployment faster and reduces compatibility issues.

3. Amazon ECS task definitions



Amazon ECS task definitions are configuration files in Amazon ECS that describe how a containerized application should run. A task definition specifies details such as the Docker image to use, CPU and memory requirements, networking settings, environment variables, and container ports. It acts like a blueprint for launching containers in ECS, allowing developers to define and replicate application setups easily across multiple environments.

4. Application Load Balancer



Application Load Balancer (ALB) is a service provided by Amazon Web Services that automatically distributes incoming application traffic across multiple targets, such as containers, instances, or IP addresses. It operates at the application layer (Layer 7) and can route traffic based on content like URL paths or hostnames. In ECS, an ALB is commonly used to balance traffic across multiple running containers, improving availability, fault tolerance, and scalability of applications.

Procedure:

<https://docs.aws.amazon.com/hands-on/latest/deploy-docker-containers/deploy-docker-containers.html>

Post-Experiments Exercise

Deploy Docker Containers on Amazon ECS (YouTube Video):

<https://www.youtube.com/watch?v=9hCZYBs0h9I>

Watch video

Extended Theory: (to be written in hand)

1. How is Amazon ECS different from AWS Elastic Beanstalk?

Results/Calculations/Observations:

Fill the following observation tables-(to be written in hand)

Amazon Aurora Vs RDS

Sr. No	Questions	Yes/No	Justify
1.	Does Amazon ECS support any other container types?		
2.	Does Amazon ECS support batch jobs?		
3.	Can user use their own Amazon Machine Image (AMI) on ECS?		

Conclusion:

1. Write what was performed in the experiment
2. Mention a few applications of what was studied.
3. Write the significance of the studied topic

References:

- [1] [Online]<https://ahireharshal.com/cloud/aws-ecs/>
[2] [Online]<https://docs.docker.com/cloud/ecs-integration/>